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Improved time complexity for spintronic oscillator ising machines compared to a popular classical optimization algorithm for the Max-Cut problem

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Abstract

Solving certain combinatorial optimization problems like Max-Cut becomes challenging once the graph size and edge connectivity increase beyond a threshold, with brute-force algorithms which solve such problems exactly on conventional digital computers having the bottleneck of exponential time complexity. Hence currently, such problems are instead solved approximately using algorithms like Goemans–Williamson (GW) algorithm, run on conventional computers with polynomial time complexity. Phase binarized oscillators (PBOs), also often known as oscillator Ising machines, have been proposed as an alternative to solve such problems. In this paper, restricting ourselves to the combinatorial optimization problem Max-Cut solved on three kinds of graphs (Mobius Ladder, random cubic, Erdős Rényi) up to 100 nodes, we empirically show that computation time/time to solution (TTS) for PBOs (captured through Kuramoto model) grows at a much lower rate (logarithmically: $\mathcal{O}(\log(N))$, with respect to graph size N) compared to GW algorithm, for which TTS increases as square of graph size ($\mathcal{O}(N^2)$). However, Kuramoto model being a physics-agnostic mathematical model, this time complexity/ TTS trend for PBOs is a general trend and is device-physics agnostic. So for more specific results, we choose spintronic oscillators, known for their high operating frequency (in GHz), and model them through Slavin's model which captures the physics of their coupled magnetization oscillation dynamics. We thereby empirically show that TTS of spintronic oscillators also grows logarithmically with graph size ($\mathcal{O}(\log(N))$), while their accuracy is comparable to that of GW. So spintronic oscillators have improved time complexity over GW algorithm. For large graphs, they are expected to compute Max-Cut values much faster than GW algorithm, as well as other oscillators operating at lower frequencies, while maintaining the same level of accuracy.

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1. Introduction

1.1. Motivation

Solving combinatorial optimization problems is useful for various practical applications: delivery scheduling, flight booking, airline crew pairing, VLSI design, optimization for machine learning problems, drug discovery, etc [1–3]. However, solving many of these problems accurately through a classical brute-force solver (which generates and tries out all possible solutions to find the correct one) on a conventional classical digital computing unit consumes exponential time. Hence, beyond a certain problem size, conventional computers cannot be used to accurately find solutions to these problems. (For example, in this paper, we are unable to find accurate solution to the combinatorial optimization problem Max-Cut on a conventional computer for some graph types with graph size above 20 nodes because the brute-force algorithm takes a very long time to run.)

In computer science, computation time that an algorithm takes is represented through its time complexity, following the big O notation [4]. In simple words, time complexity is a measure of the rate at which the computation time grows with problem size. The time complexity of the brute-force algorithms to solve the aforementioned combinatorial optimization problems is exponential, leading to their lack of feasibility.

The current popular technique to solve such problems is using classical approximation algorithms, implemented on conventional computers only. These algorithms do not yield accurate solutions but yield close-to-accurate solutions, and their advantage is their computation time/time to solution (TTS) grows polynomially with problem size as opposed to exponentially (i.e., they have polynomial time complexity). One such popular algorithm is the semi-definite-programming (SDP)-based Goemans–Williamson (GW) algorithm (briefly explained later in this paper), which offers an approximation ratio (AR) guarantee ≈ 0.878 or higher for the Max-Cut problem [5, 6].

Various emerging computing hardware systems have been proposed and experimentally demonstrated recently, targeted to solve these problems more efficiently. Among them, hardware systems that convert the problem to its corresponding Ising Hamiltonian [7] and then find the ground state of the Hamiltonian are known as Ising machines [8]. Quadratic Unconstrained Binary Optimization (QUBO) is also often used for the same purpose, and it has been shown that QUBO and the Ising model are equivalent via a linear transformation of variables [9].

Ising machines have been implemented through various emerging technologies [8, 10, 11]. Out of them, one popular

class of technologies is the phase-binarized oscillator (PBO) system [12], also popularly known as oscillator Ising machines (OIM) [13, 14] or oscillatory neural networks [15]. PBOs/OIMs have been reported using a wide range of device technologies: electronic inductor-capacitor (LC) oscillator [16], electronic ring oscillator [3, 17, 18], electronic Schmitt Trigger circuit [15, 19], metal–insulator–transition device [20, 21], etc.

Spintronic oscillators have also been proposed recently for Ising computing [9, 22–24] since they offer various advantages in this regard. They can be coupled through various means like dipole coupling [25–27], spin–wave coupling [9, 22], and electrical coupling [28, 29]. They operate at a very high frequency (in GHz), leading to faster solutions compared to other aforementioned oscillator technologies which operate in several MHz [3, 15, 16] as opposed to GHz. Spin oscillators are also known to consume very low energy.

1.2. Our contribution

The main competition for spintronic-oscillator-based Ising machines, or for other OIMs for that matter, is with classical approximation algorithms like GW algorithm meant for solving the same problems (like Max-Cut) on a conventional computer. So, for graphs of different edge connectivity degrees (hence different difficulty levels), it is important to compare how computation time or TTS of OIMs and GW algorithm grow with graph size/number of nodes (N), along with comparing the accuracy of OIMs and GW algorithm. We carry out this time complexity comparison in this paper.

It is to be noted that since Ising machines are heuristic solvers, obtaining their time complexity analytically is difficult [8]. So we carry out this time complexity comparison empirically, much like it has been done for optical coherent Ising machines (CIMs) by Haribara and Yamamoto [10] and Hamerley et al [11].

Restricting ourselves to the Max-Cut problem, we solve for three kinds of graphs (unweighted Mobius Ladder (ML) [30], unweighted random cubic (RC) [31], unweighted Erdős Rényi (ER) [32]) up to 100 graph nodes ($N = 100$) by numerically simulating a system of generalized PBOs, represented through Kuramoto model [13, 14, 33], as well as the GW algorithm. We show that PBOs offer the same level of accuracy as GW algorithm, but computation time/TTS of GW algorithm increases as square of graph size ($\mathcal{O}(N^2)$), following the big O notation [4, 6], and TTS of generalized PBOs increases only logarithmically with graph size ($\mathcal{O}(\log(N))$). This shows that PBOs have improved time complexity over GW algorithm in terms of solving Max-Cut on large graphs. Incorporation of noise in the model does not affect the accuracy or TTS much.

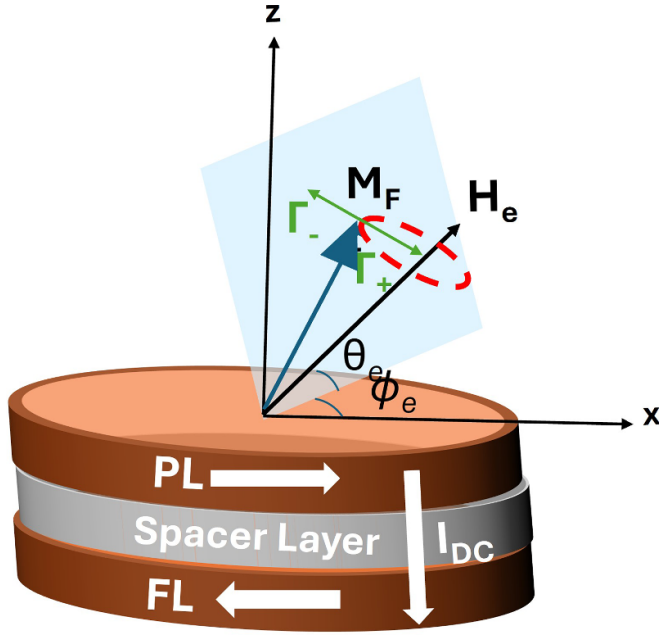


Figure 1. Schematic of the spin transfer torque nano oscillator (STNO) captured by Slavin's model [34] in this work. PL: pinned layer (the layer in which magnetic moment is fixed). FL: free layer (the layer in which the magnetic moment M_F precesses about an axis at steady state: oscillation. H_e : applied magnetic field, and θ_e, ϕ_e : its direction cosines. I_{DC} is the current which generates spin transfer torque and triggers magnetic oscillation. Γ_+ and Γ_- are defined in the text [35].

However, Kuramoto model is a physics-agnostic mathematical model which describes coupled oscillator dynamics without taking into account their exact coupling physics. So, the aforementioned time complexity/TTS advantage predicted on the basis of Kuramoto model will be applicable to physical oscillators provided that their dynamics is consistent with the Kuramoto model. Based on by how much the dynamics of the actual oscillators deviates from the Kuramoto model, the above TTS trend will also change.

To get more specific results, we here choose spintronic oscillators among all possible oscillator technologies. We make this particular choice because of the aforementioned advantages of spintronic oscillators (including high speed, so lower TTS than other oscillators). We model a system of coupled spintronic oscillators represented through Slavin's model, which takes care of their coupled magnetization dynamics [34, 35].

We show that the accuracy of spintronic oscillators is comparable to that of generalized PBOs and GW algorithm. We also show that just like generalized PBOs, TTS of spintronic oscillators increases only logarithmically with graph size ($\mathcal{O}(\log(N))$) for all three kinds of graphs considered here (N up to 100 considered). Thus spintronic oscillators have an improved time complexity and can provide a much faster alternative compared to GW algorithm for solving Max-Cut on large graphs.

It is to be noted here that a previous work by Albertsson *et al* [35] reports Max-Cut solutions on quite large graphs using the

Slavin's model of spin oscillators. But comparison between TTS of spin oscillators and that of any conventional state-of-the-art optimization algorithm (like GW), thereby showing this kind of time complexity advantage of spintronic oscillators, has not been reported before, to the best of our knowledge. It is also to be noted that the previously mentioned report by Haribara and Yamamoto shows that optical CIMs also have similar kind of time complexity advantage compared to GW algorithm as spintronic oscillators do [10]. However, CIMs have major disadvantages like the use of a long fibre ring cavity and a power-hungry field-programmable gate array to implement coupling, which is not the case with spintronic oscillators [21].

In section 2 of this paper, we discuss our oscillator modeling methods. In section 3, the Max-Cut problem is formally defined, and the graph types used here (ML, RC, ER) and the GW algorithm are introduced. Section 4 shows the comparison results for generalized PBOs, spintronic oscillators, and GW algorithm, with respect to Max-Cut solved on those graph instances. In section 5, we make further interpretations from the comparison results presented in section 4 and conclude the paper.

2. Modelling of OIMs

2.1. Kuramoto model for generalized PBOs

We use the Kuramoto model here to model the dynamics of PBOs in a generalized way. In the next sub-section, we use a model more specific to spintronic oscillators.

According to the Kuramoto model, the net energy of a system of coupled oscillators under sub-harmonic injection locking (SHIL) is given by:

$$E = -\sum_{j'=1}^n \sum_{j=1, j < j'}^n K_{jj'} \cos(\bar{\phi}_j - \bar{\phi}_{j'}) - K_s \sum_{j=1}^n \cos(2\bar{\phi}_j), \quad (1)$$

where $\bar{\phi}_j$: phase of oscillator j , K_s : anisotropy constant (leads to SHIL), and $K_{jj'}$: coupling constant between oscillator j and oscillator j' [28]. SHIL contributes to phase binarization of the oscillators, as discussed in detail subsequently in section 5.1 and also in reports by Wang *et al* [13, 14].

Phases of oscillators evolve over time (t) as follows:

$$\frac{d\bar{\phi}_j}{dt} = \omega_j - \omega^* - \sum_{j' (j' \neq j)}^N K_{jj'} \sin(\bar{\phi}_j - \bar{\phi}_{j'}) - 2K_s \sin(2\bar{\phi}_j), \quad (2)$$

where ω_j : natural frequency of oscillator j , ω^* : average natural frequency of all the oscillators [28]. When the natural frequencies of the oscillators are equal (as in the cases considered in this paper), $\frac{d\bar{\phi}_j}{dt} = -\frac{\partial E}{\partial \bar{\phi}_j}$, and thus the phases evolve such that the net energy of the system E (as per equation (1)) goes to a minimum, as expected in Ising computing [28].

We solve equation (2) numerically using Runge Kutta method, with each time-step $\Delta t = 5 \times 10^{-5}$ s, and number of

time-steps = 10^6 . We use the following values in the Kuramoto model: $\omega_j = 0.1$ radians s^{-1} for all oscillators labelled j from 1 to N (hence, $\omega^* = 0.1$ radians s^{-1}), and $K_s = 0.5$ radians s^{-1} . With respect to each graph instance used (all unweighted in nature) for solving Max-Cut, we take $K_{j,j'} = 1$ radian s^{-1} (equation (2)) corresponding to all existing edges between node j and node j' , and $K_{j,j'} = 0$ whenever no edge exists between nodes labeled j and j' .

It is to be noted here that given the Kuramoto model is a physics-agnostic model, the choice of frequency value is arbitrary. Hence, TTS obtained from the Kuramoto model will be reported in terms of number of time steps in this paper. The exact value of the time step in seconds can vary based on the operating frequency of the oscillator.

2.2. Slavin's model of spin oscillators

In the model proposed by Slavin and Tiberkevich [34] for spintronic auto-oscillators as shown in figure 1, each oscillator j is represented by its complex spin-oscillation amplitude c_j and dimensionless spin wave power $p_j = |c_j|^2$ which is proportional to the experimentally measured microwave power of the spin oscillator. The equations used to model the array of coupled spin oscillators, as per this Slavin's model, are given by:

$$\frac{dc_j}{dt} = -i\omega_j(p_j)c_j - \Gamma_{+j}(p_j)c_j + \Gamma_{-j}(p_j)c_j + K_e e^{-i\omega_e t} c_j^* + \sum_{j'} \Omega_{j,j'} e^{i\beta_{j,j'}} c_{j'}. \quad (3)$$

Equation (3) can be decoupled into two equations for power $p_j(t)$ and phase $\hat{\phi}_j(t)$ of c_j as:

$$\frac{dp_j}{dt} = -2p_j[\Gamma_{+j}(p_j) - \Gamma_{-j}(p_j)] + 2K_e p_j \cos(\omega_e t + 2\hat{\phi}_j) + 2 \sum_{j,j'} \Omega_{j,j'} \sqrt{p_j p_{j'}} \cos(\hat{\phi}_j - \hat{\phi}_{j'} - \beta_{j,j'}) \quad (4)$$

$$\frac{d\hat{\phi}_j}{dt} = -\omega_j(p_j) - K_e \sin(\omega_e t + 2\hat{\phi}_j) + \sum_{j,j'} \Omega_{j,j'} \sqrt{p_{j'}/p_j} \sin(\hat{\phi}_{j'} - \hat{\phi}_j + \beta_{j,j'}), \quad (5)$$

where $\omega_j(p_j) = \omega_0 + \mathcal{N}p_j$ is the precession term which represents the operating frequency, $\mathcal{N}/2\pi \approx -3.44$ GHz is the nonlinear frequency shift coefficient, $\Gamma_{+j}(p_j) = \Gamma_G(1 + Qp_j)$ is the damping term with $Q \approx 2.66$ and $\Gamma_G/2\pi \approx 252$ MHz, $\Gamma_{-j}(p_j) = \sigma I_{dc}(1 - p_j)$ is the anti-damping term due to the spin polarised direct current I_{dc} and $\sigma \approx 2.5$ GHz mA $^{-1}$. The above parameter values are taken from the report by Albertsson *et al* [35] and that by Chen *et al* [36]. Here, $\Omega_{j,j'}$: coupling constant between oscillators j and j' , $\beta_{j,j'}$: phase constant between oscillators j and j' . For SHIL, an external RF current of frequency ω_e and coupling strength K_e is injected into the spin oscillators.

In our work, we solve the coupled differential equations (4) and (5) numerically, with a time range from 0 to 200 ns

divided into 100000 intervals of equal size. We use $\beta_{j,j'} = -1.6\pi$, $K_e = 25$ MHz, $\omega_0/2\pi = 4.2$ GHz, and $I_{dc} = 1.5$ mA. With respect to the graph instances used here (all unweighted in nature) for solving Max-Cut, we take $\Omega_{j,j'} = 25$ MHz (equations (4) and (5)) in Slavin's model corresponding to all existing edges between node j and node j' , and $\Omega_{j,j'} = 0$ whenever no edge exists between nodes labeled j and j' .

It is to be noted that no noise term is included in the aforementioned equations for Kuramoto model and Slavin's model. So the results presented in section 4 (figures 4–7) for Kuramoto model do not incorporate noise. The method to incorporate noise and the impact of noise on OIM's performance are presented in section 5.2.

3. Max-Cut, graph instances, and GW algorithm

3.1. The Max-Cut problem and connection with oscillator models

The Max-Cut problem for an unweighted graph (we only consider unweighted graphs in this work) is defined here formally. Given an undirected weighted graph $\mathcal{G} = (V, E)$, where V and E are the graph's vertex and edge sets respectively, and weights w_{ij} , the objective function for Max-Cut can be expressed as:

$$\begin{aligned} \max \quad & \frac{1}{2} \sum_{(j,j') \in E} w_{j,j'} (1 - x_j x_{j'}) \\ \text{subject to} \quad & x_j \in \{-1, 1\} \quad , \text{ for } i \in V. \end{aligned} \quad (6)$$

For unweighted graphs, $w_{j,j'} = 1$ if an edge connects node j and j' in the graph, otherwise 0. The Max-Cut problem is about partitioning the graph (assigning a value, 1 or -1 , to x_j for every node j) such that the value of the objective function is maximized.

Maximizing this objective function of the Max-Cut problem can be directly mapped to minimizing the energy/Hamiltonian in the Ising model of spins, as given by:

$$H = -J_{j,j'} S_j S_{j'}, \quad (7)$$

where spins S_j and $S_{j'}$ take values of 1 or -1 , and corresponding to positive weights. Corresponding to positive weights $w_{j,j'}$ of edges connecting nodes j and j' of the graph, $J_{j,j'}$ in the Ising model above (equation (7)) take negative values, with the magnitude of $J_{j,j'}$ being proportional to $w_{j,j'}$.

The connection between Ising model of spins and Kuramoto model of oscillators is derived in detail in the report by McGoldrick [28]. As explained there, $K_{j,j'}$ of Kuramoto model (equation (2)) corresponds to weight $w_{j,j'}$ of the graph and $J_{j,j'}$ of the Ising model of spins. Similarly, $\Omega_{j,j'}$ in Slavin's model (equations (4) and (5)) is proportional to $w_{j,j'}$ of the graph and $J_{j,j'}$ of the Ising model of spins. As a result, for all the unweighted graphs we have explored in this paper, if two nodes j and j' have an edge between them ($w_{j,j'} = 1$), then $K_{j,j'} = 1$ radian s^{-1} in Kuramoto model and $\Omega_{j,j'} = 25$ MHz in Slavin's model, as mentioned earlier. If two nodes j and j' do not have an edge between them ($w_{j,j'} = 0$), then $K_{j,j'} = 0$

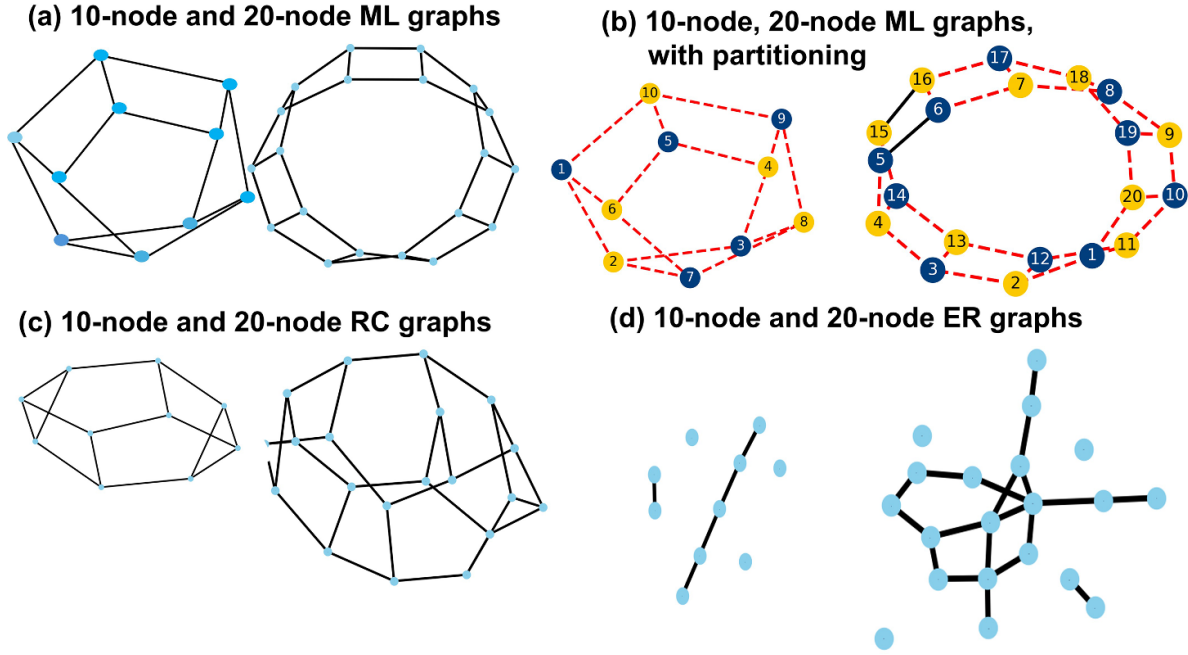


Figure 2. (a) 10-node and 20-node Mobius ladder (ML) graphs are shown without showing any partitioning. (b) Partitioning of the graphs in (a) such that the corresponding score for the Max-Cut problem is maximized. Nodes belonging to one partition are shown as dark blue circles, nodes belonging to the other partition are shown as yellow circles, and the edges that contribute to the final score that is to be maximized are shown as red dashed lines. (c) 10-node and 20-node instances of the random cubic (RC) graph used in this work (d) 10-node and 20-node instances of the Erdős Rényi (ER) graph used in this work. We consider ER graphs $G(N, 0.1)$ throughout this work, where N : number of nodes, 0.1 is the probability for an edge to occur (more details in text).

radian s^{-1} in Kuramoto model and $\Omega_{jj'} = 0$ in Slavín's model, as mentioned earlier.

3.2. Types of graphs used

Mobius ladder graph: A Mobius Ladder (ML) graph (shown in figure 2(a)) of graph size N (N is a positive even integer) is an undirected unweighted 3-regular (also known as cubic) graph with N nodes, for which every node i has edges with only three other nodes [30]:

$$\begin{cases} i-1, i+1, \left(\frac{N}{2}\right) + i, & \text{if } 2 \leq i \leq \frac{N}{2} \\ i-1, i+1, i - \left(\frac{N}{2}\right), & \text{if } \left(\frac{N}{2}\right) + 1 \leq i < N \\ n, i+1, \left(\frac{N}{2}\right) + i & \text{if } i = 1 \\ i-1, 1, i - \left(\frac{N}{2}\right) & \text{if } i = N. \end{cases}$$

With all other nodes, node i does not have edges. Figure 2(a) shows Mobius ladder graphs of graph size (number of nodes) 10 and 20. Correct partitioning of these graphs, yielding the correct Max-Cut score, is shown in figure 2(b). In this work, we have used Mobius Ladder graphs with $N = 10, 20, 30, 100$.

RC graph: Any RC graph we use here is an undirected unweighted 3-regular graph where each node is randomly connected to three other nodes [31]. In our paper, for graph size $N = 10, 20, 30, 100$, we generate a RC graph using the NetworkX Python package [31, 37, 38]. The specific 10-node and 20-node RC graph instances we generate for this work are shown in figure 2(c).

ER graph: Any ER graph we use here is given by $G(N, p)$ and is an undirected unweighted graph of N nodes, where the probability of any edge between two nodes to exist (also known as edge probability) is independent of any other edge to exist and is given by p ($0 \leq p \leq 1$) [32]. Thus, the expected number of nodes for this graph is $\binom{pN}{k}$. In our paper, for graph size $N = 10, 20, 30, 100$, we generate an ER graph given by $G(N, 0.1)$. The specific 10-node and 20-node ER graph instances we generate for this work are shown in figure 2(d).

3.3. Classical GW algorithm

Classical GW algorithm optimizes the Max-Cut objective function (equation (6)) through approximation (as described below). Since GW algorithm can be implemented on conventional digital computers, it is very commonly used for solving the Max-Cut problem on large graphs as mentioned before [5]. In the GW formulation, the MaxCut objective function is relaxed and reformulated as:

$$\begin{aligned} \max \quad & \frac{1}{2} \sum_{(i,j) \in E} w_{ij} (1 - X_{ij}) \\ \text{subject to} \quad & \vec{a}^T X \vec{a} \geq 0, \quad \forall \vec{a} \in \mathbb{R}^N, \\ & X_{ii} = 1, \quad \text{for } i \in V \\ & X \in \mathbb{R}^{N \times N} \end{aligned} \quad (8)$$

where $X_{i,j} = \vec{x}_i \cdot \vec{x}_j$, $(\cdot)$ is the inner product and $|V| = N$ (size of the graph). The binary variables x in equation (6) are replaced by vectors $\vec{x}$ on an N -dimensional unit sphere i.e. $\vec{x} \in \mathbb{S}^N$. This

is known as SDP relaxation. Solving for X yields the optimal vector $\vec{x}_{\text{opt}}$ for every vertex. We have used here the CVXOPT [39] and SciPy [40] python packages to solve for $\vec{x}_{\text{opt}}$.

Next, we draw a random hyperplane passing through the origin of $\mathbb{S}^N$ with the corresponding normal vector to the plane being $\vec{r}$. The rationale behind this step is to partition the vectors into two sub-spaces where vectors in one subspace are assigned the value -1 and in the other sub-space 1 , after investigating the sign of $(\vec{x}_{\text{opt}} \cdot \vec{r})$. If this inner product is positive (negative), then $\vec{x}_{\text{opt}}$ is mapped to $+1(-1)$. This part of the algorithm is the measurement step. We take 1024 such random normal vectors $\vec{r}$ and thus obtain 1024 cut values, from which the highest cut value is obtained and used as a benchmark for the other solvers, as described in the next section.

4. Comparing performance of OIMs with GW algorithm

4.1. Accuracy of solution

For each graph instance described in the previous section, we solve equations (4) and (5) numerically 100 times (100 trials) for Slavin's model of spin oscillators and solve equation (2) numerically 100 times (100 trials) for the Kuramoto model of generalized PBOs, with different initial phase values each time, to obtain the phase dynamics of the oscillators. The final stable solution that the oscillator phases attain, after possibly crossing some meta-stable states corresponding to sub-optimal solutions, is recorded for each graph instance. Such phase evolution corresponding to the 20-node ML graph considered here is shown in figure 3.

From each stable-state phase solution, the corresponding graph partitioning and cut score are obtained as follows (such calculation for 20-node ML graph is shown in figure 3). With respect to oscillator 1 (reference oscillator), the phase difference of each other oscillator is rounded off to an integral multiple of π . If the integer is an even number, then the node corresponding to that oscillator belongs to the same partition as the node corresponding to the reference oscillator (shown with blue circles in the ML graph of figure 3). If the integer is an odd number, then the node corresponding to that oscillator belongs to the opposite partition as the node corresponding to the reference oscillator (shown with yellow circles in the ML graph of figure 3). Thus values x_i are assigned to the nodes (opposite partitions: opposite signs for x_i), and using equation (6), the value of the objective function is obtained as the cut score. Thus, for each graph instance, we do 100 trials and hence obtain 100 cut values both for the Kuramoto model of generalized PBOs and Slavin's model of spin oscillators.

For benchmarking of these OIM-based solutions for graphs upto the size of 20 nodes, the classical accurate brute-force solver is used. Beyond that size, the classical brute-force solver does not yield the optimal solution because computation time grows exponentially, as mentioned before. Nonetheless, Max-Cut on ML graphs is polynomial-time solvable [41]. For a ML graph with N nodes, if $\frac{N}{2}$ is odd, then the Max-Cut score is $3\frac{N}{2}$; if $\frac{N}{2}$ is even, then the Max-Cut score is $\frac{3N}{2} - 2$. So we use this formula to obtain the correct Max-Cut score for benchmarking

in the case of ML graphs upto 100 nodes. However, for RC and ER graphs for number of nodes $N = 30, 40, 100$, the highest cut of the 1024 cut values obtained through GW algorithm (as described in the previous section) for a given graph instance is considered the best known Max-Cut solution for that graph instance, in the absence of any accurate solution from the brute force solver.

For each graph instance, the average/ mean of all the cut values obtained for a given model (Kuramoto model: 100 cut values, Slavin's model: 100 cut values) is divided by the correct/ best known Max-Cut score (calculated as described above) to obtain the mean AR for the OIM. Mean ARs of OIMs for the three types of graphs used here, with graph sizes ranging from 10 to 100 nodes, are shown in figure 4. Standard deviation across those cut values, divided by the correct/ best known Max-Cut score, is plotted as an error bar in these plots (figure 4). The highest cut value out of all the cut values obtained for a graph instance is divided by the correct/ best known Max-Cut score to obtain the best case AR. Best case ARs of OIMs for the same graph instances are plotted in figure 5. It is to be noted that no thermal noise has been incorporated in these simulations. The role of thermal noise is studied subsequently in section 5.2.

Figures 4 and 5 show that the mean AR and best case AR of spintronic oscillators, as obtained from Slavin's model, closely match with that of generalized PBOs (Kuramoto model) for almost all graph instances. This is expected because previous studies have shown that the dynamics of coupled spintronic oscillators indeed follows closely the dynamics of generalized coupled oscillators, as predicted by the Kuramoto model [25, 42, 43].

For all ML graph instances, best case AR is the ratio of highest cut value obtained from OIM to the correct Max Cut value (as mentioned earlier). Since the best case AR for generalized PBOs (Kuramoto model) is 1 for all considered ML instances and that for spintronic oscillators (Slavin's model) is 1 for almost all ML instances, it can be inferred that out of 100 trials, at least in some trials OIMs can yield correct Max-Cut scores for almost all ML graph instances. However, the performance of these OIMs is slightly sub-optimal for RC and ER graphs. This is expected because ML graphs have very low edge connectivity and have a definite pattern, and hence solving Max-Cut on them is not really NP-Hard (as mentioned earlier) [41]. This is not the case with RC and ER graphs. Nonetheless, since mean AR and best case AR are defined as the ratios of mean and highest cut values from OIM to highest cut value obtained from GW algorithm for RC and ER graphs with size more than 20 nodes (as mentioned earlier), the fact that mean AR and best case AR values for both generalized PBOs and spintronic oscillators are close to 1 all the way up to graphs with 100 nodes shows that the accuracy of OIMs is comparable to that of GW.

4.2. Computation time/TTS

To obtain TTS on a given graph instance for the Kuramoto model of generalized PBOs and Slavin's model of spin oscillators, we consider all 100 trials and calculate the average time

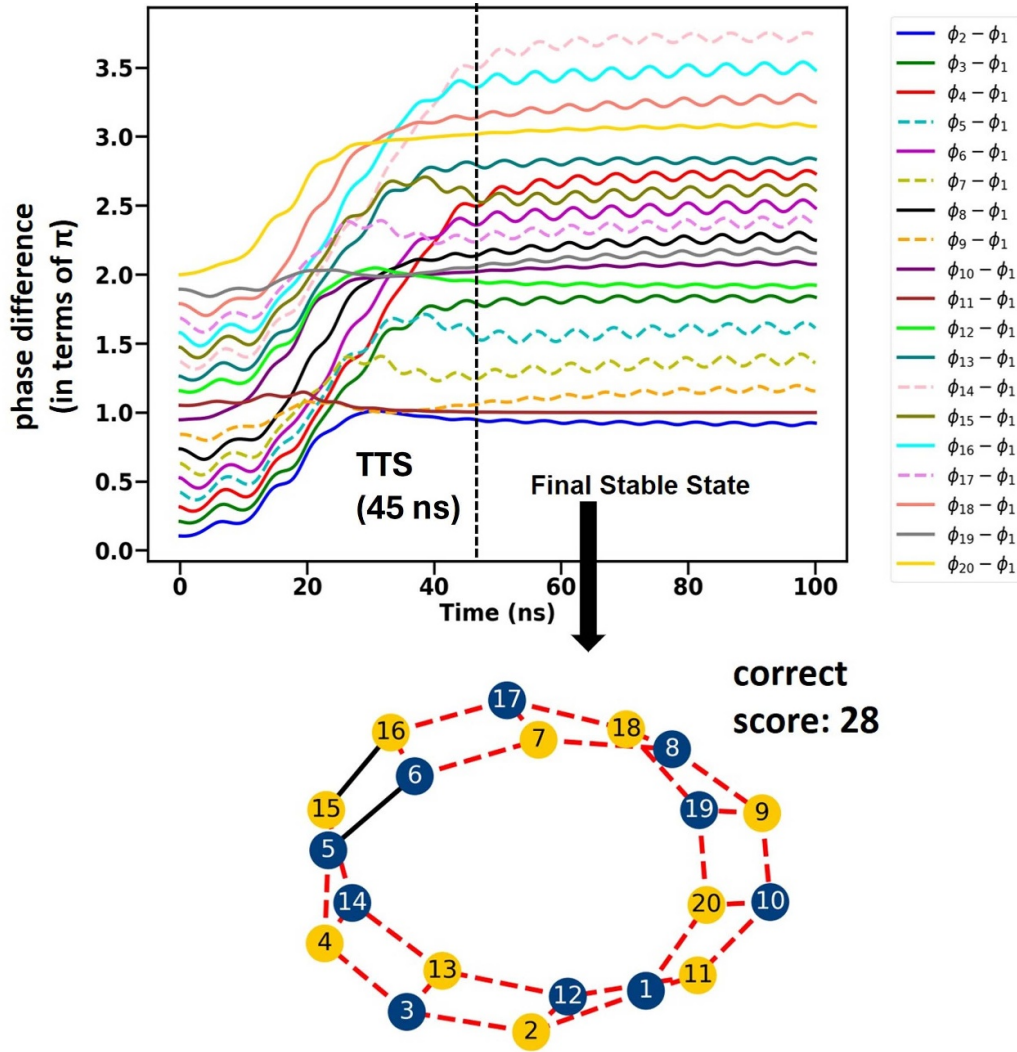


Figure 3. Evolution of the phases of spintronic oscillators, modelled by Slavin's model, over time corresponding to the 20-node Mobius ladder (ML) graph shown in figure 2(a). Difference between the phase ϕ of every other oscillator (numbered 2 to 20) and the phase of oscillator 1 (ϕ_1), normalized by π , has been plotted. Based on the phase differences in the final stable state, the partitioning is obtained (as described in text) which matches with that shown in figure 2(b). The time at which the final steady state is obtained is reported as time to solution (TTS).

(in nanoseconds for spin oscillators, in number of time steps for generalized PBOs) when the phases of the oscillators reach a stable state (after crossing meta-stable states that may occur) and do not change further. For example, in the phase evolution shown for spintronic oscillators in the case of 20-node ML graph (figure 3), the time when the final stable state is reached (45 ns) is considered the TTS for that particular trial. The average across 100 such trials is taken to calculate TTS for spintronic oscillators on the 10-node RC graph instance considered.

To calculate TTS for the GW algorithm, we measure the time taken for solving the SDP relaxation and one random hyperplane measurement on a conventional digital computer with the following specifications: 11th Gen Intel(R) Core(TM) i5-1135 processor and 16GB memory. We measure 1000 such times by repeating the above process for a given graph instance and obtain the mean time in milliseconds.

For the three kinds of graphs (ML, RC, ER) considered here, TTS for graph instances with 10 nodes is listed for generalized PBOs, spintronic oscillators, and GW algorithm in table 1. In figure 6, for all these graph types all the way up to 100 nodes, TTS is plotted as a function of number of graph nodes (N) in a normalized fashion. For PBOs, the actual TTS (in number of steps) is normalized by TTS for PBOs (in number of steps) for the graph instance of the same graph type with 10 nodes (as listed in table 1). Similarly, for spintronic oscillators, the actual TTS (in ns) is normalized by TTS for spin oscillators (in ns) for the graph instance of the same graph type with 10 nodes (as listed in table 1). For GW algorithm, the actual TTS (in ms) is normalized by TTS for spin oscillators (in ms) for the graph instance of the same graph type with 10 nodes (as listed in table 1).

From figure 6, it is clear that while TTS for GW algorithm grows very rapidly with graph size (N), TTS for OIMs (both

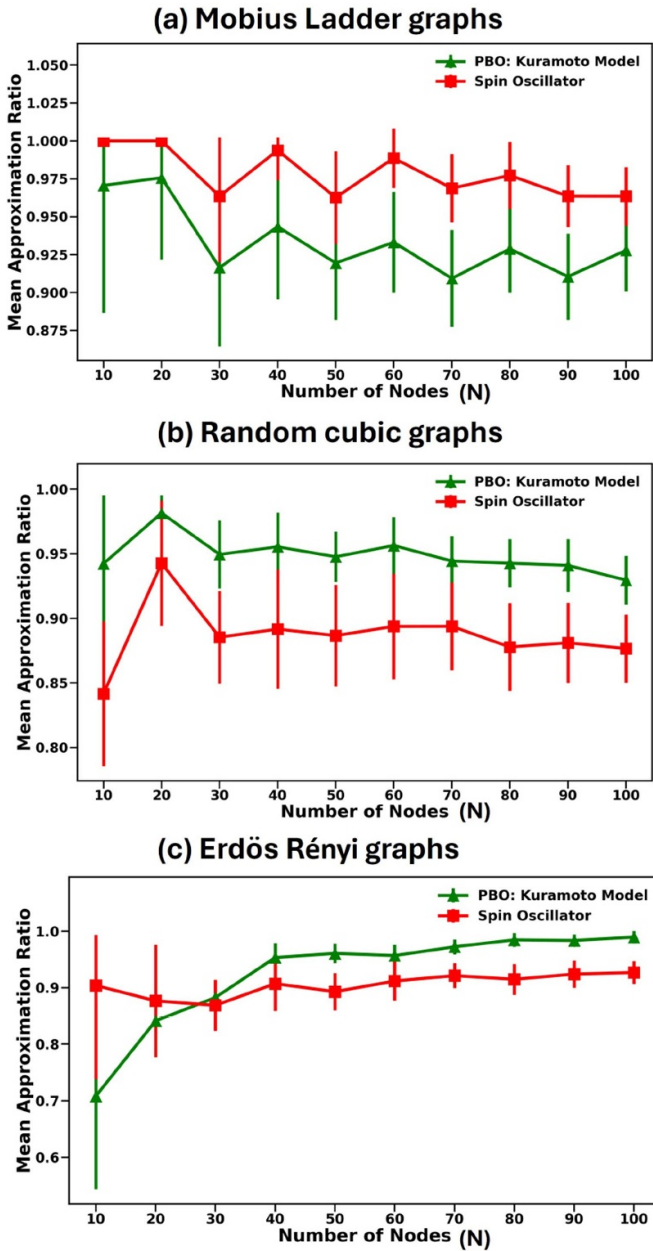


Figure 4. Mean approximation ratio vs size of the graph (in terms of number of nodes N) for both generalized PBOs (Kuramoto model) (green plot) and spintronic oscillators (Slavin's model) (red plot) and three kinds of graphs: (a) Mobius Ladder, (b) random cubic, (c) and Erdős Rényi. Standard deviation of cut values (divided by best known Max-Cut value) is shown as an error bar around the mean value. The method to calculate mean approximation ratio and standard deviation is described in the text. No thermal noise has been incorporated in either Kuramoto model or Slavin's model to obtain this plot.

generalized PBOs and spintronic oscillators) grows at a much lower pace with graph size. For a more quantitative interpretation, we fit functions of N on these TTS vs N plots. We observe that for the same range of error in fitting, TTS varies as $\mathcal{O}(N^2)$ (following the big O notation and thereby ignoring coefficients and lower order terms in the fitting function [4]) with graph size N for GW algorithm implemented on

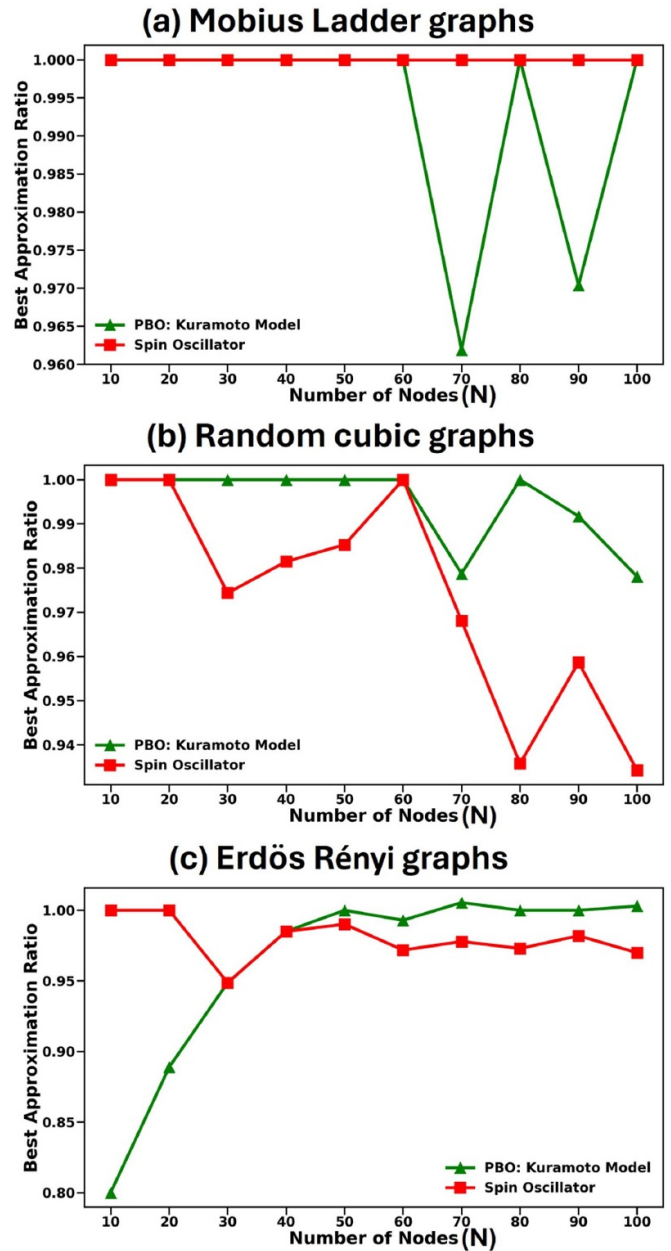


Figure 5. Best case approximation ratio vs size of the graph (in terms of number of nodes N) for both generalized PBOs (Kuramoto model) (green plot) and spintronic oscillators (Slavin's model) (red plot) and three kinds of graphs: (a) Mobius Ladder, (b) random cubic, (c) and Erdős Rényi. The method to calculate best case approximation ratio is described in the text. No thermal noise has been incorporated in either Kuramoto model or Slavin's model to obtain this plot.

all the three types of graphs considered here (ML, RC, ER). On the other hand, TTS for both generalized PBOs (Kuramoto model) and spintronic oscillators (Slavin's model) varies only as $(\mathcal{O}(\log(N)))$ with graph size N (again following the big O notation and thereby ignoring coefficients and lower order terms in the fitting functions [4]). This logarithmic growth is more apparent when TTS is plotted in linear scale and number of nodes is plotted in log scale, as shown in figure 7.

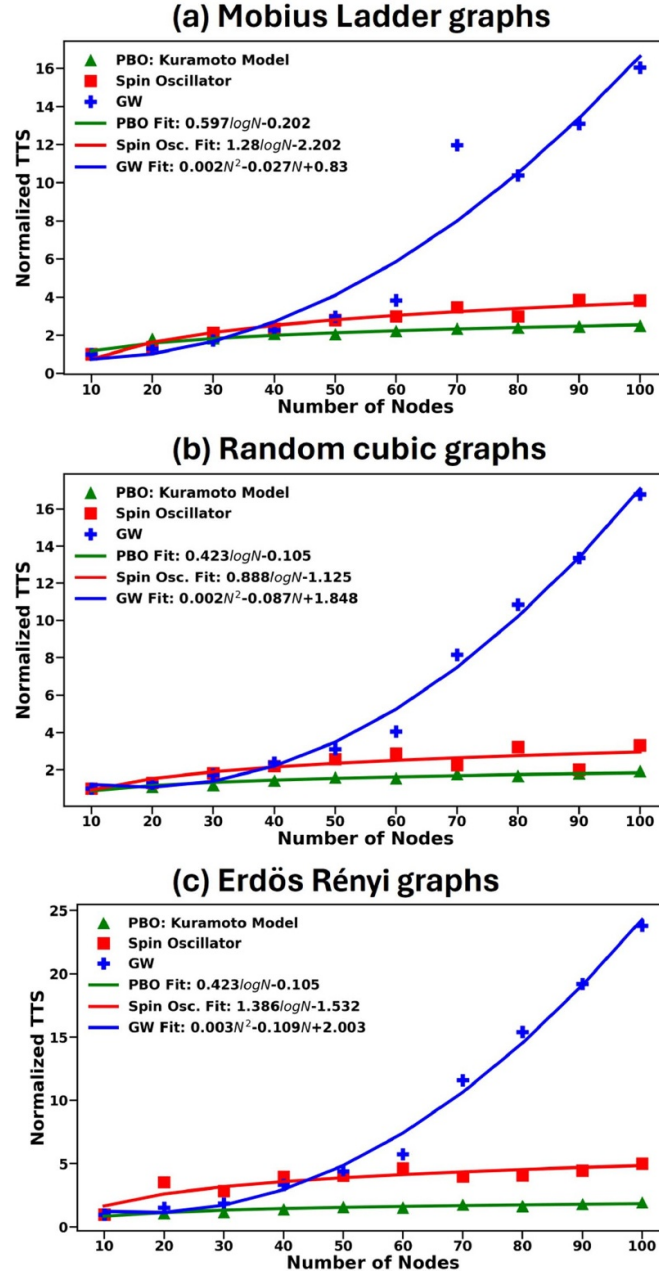


Figure 6. Time to solution (TTS), normalized by TTS for the same method for a graph instance of the same graph type with 10 nodes, vs size of the graph (in terms of number of nodes N , plotted on linear scale) for generalized PBOs (Kuramoto model) (green plot), spintronic oscillators (Slavin's model) (red plot), and GW algorithm (blue plot) for three kinds of graphs: (a) Mobius Ladder, (b) random cubic, (c) and Erdős Rényi. The method to calculate TTS and normalize it is described in the text. Best possible fits for the obtained curves, as functions of number of nodes N , are also shown. No thermal noise has been incorporated in either Kuramoto model or Slavin's model to obtain this plot.

The above finding is the key result of the paper since it shows that oscillators in general (as long as their dynamics is consistent with Kuramoto model), and spintronic oscillators more specifically, have a major advantage in terms of speed compared to classical approximation algorithms currently used to solve the same kind of optimization problems like Max-Cut (like GW algorithm) when the graphs become large in size.

5. Discussion

5.1. Role of SHIL

In the Kuramoto model which we have used in this paper to obtain the dynamics of generalized oscillators, SHIL is included through the $2K_j \sin(2\bar{\phi}_j)$ term of equation (2). Here, we have studied the contribution of SHIL towards phase

Table 1. Time to solution (TTS) for solving Max-Cut on graphs of 10 nodes using generalized PBOs (Kuramoto model), spintronic oscillators (Slavin's model), and GW algorithm. Parameter values used, value of each time step for PBOs, and specifications of the conventional computer on which GW algorithm is run for this work are mentioned in the text.

Graph instance	TTS for Generalized PBOs	TTS for Spin oscillators	TTS for GW algorithm
10-node Mobius Ladder	1.34×10^5 time steps	22.05 ns	1.64 ms
10-node random cubic	2.52×10^5 time steps	30.8 ns	1.64 ms
10-node Erdős Rényi	2.46×10^5 time steps	24.75 ns	1.57 ms

binarization of the oscillators (figure 8). In the absence of the SHIL term ($K_s = 0$ in equation (2)), final steady state phase differences of the oscillators with reference to the reference oscillator, after normalization by π , take all possible values between 0 and 2. Any value above 2 has been mapped to the $[0, 2]$ interval because of the 2π periodicity of phase. After rounding off all the phase values between 0 and 0.5, and also between 1.5 and 2 (shown by green dots), to 0, and after rounding off all the phase values between 0.5 and 1.5 (shown by red dots) to 1, the final cut values are obtained. Graph instances of all three graph types explored in the paper (Mobius ladder (ML), RC, and ER) are shown in this figure: figures 8(a), (c) and (e). On the other hand, when the SHIL term is included ($K_s = 0.5 \text{ radian s}^{-1}$, as in the original manuscript), the phase difference for any oscillator with the reference oscillator clearly binarizes to 0 (also same as 2π), given by green dot, or π , given by red dot: figures 8(b), (d) and (f). Each dot in figure 8 corresponds to an individual oscillator; as a result, as the graph size increases, there are more dots since there are more oscillators. Our analysis here shows that SHIL plays a key role in phase binarization as per the Kuramoto model of oscillators, which agrees with the findings reported by Wang *et al* [13, 14].

5.2. Role of thermal noise

As mentioned earlier, the results presented in section 4 (figures 4–7) do not include the effect of thermal noise in either the Kuramoto model or Slavin's model of oscillators. Here, we incorporate thermal noise in the models as described below and study its effect on the OIM's performance.

Following the method used by Wang *et al* [13, 14] to incorporate thermal noise in the Kuramoto model of oscillators, we have generated at every time step numbers/ sampled from a standard normal distribution $\mathcal{N}(0, 1)$. We have then multiplied this random number with a constant $k_n = 10$ to obtain noise magnitude of similar proportion as that of the coupling terms in the Kuramoto model. This resultant noise term $\xi(t) = k_n \mathcal{N}(0, 1)$ is added to the right hand side of equation (2), and then the oscillator phase ϕ_j has been solved for numerically like before for the same graph instances with other parameter values same as before.

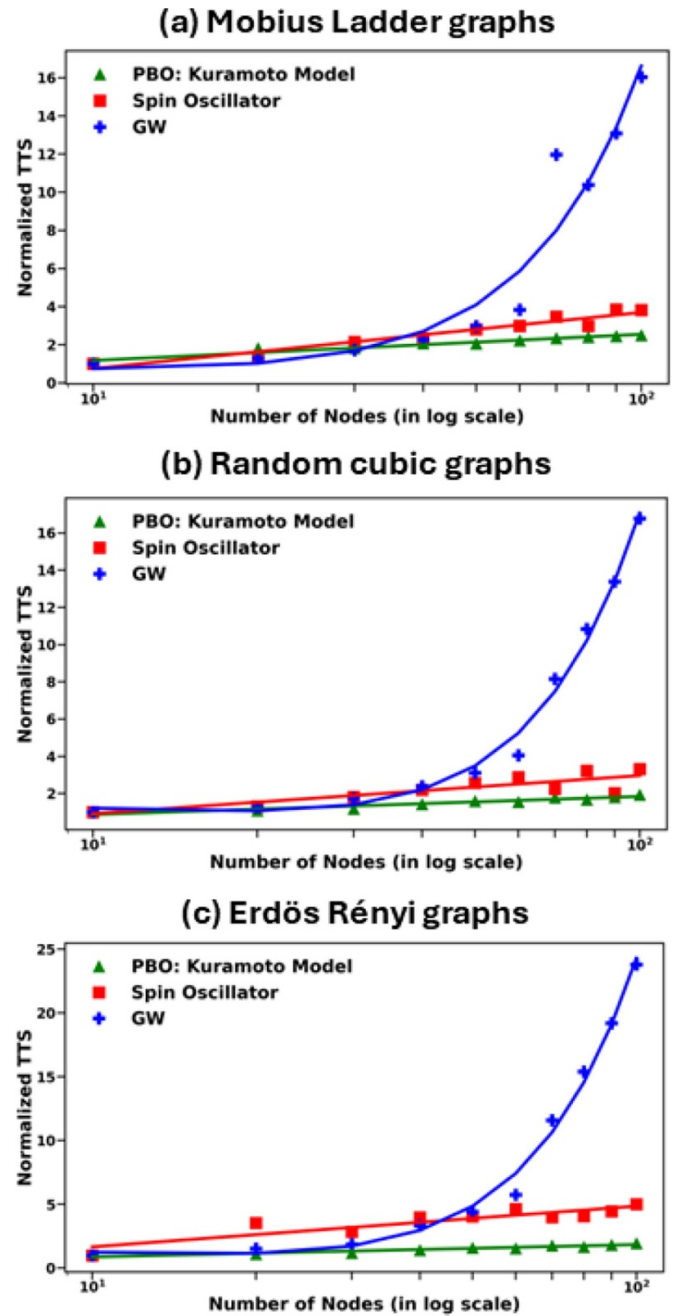


Figure 7. Time to solution (TTS), normalized by TTS for the same method for a graph instance of the same graph type with 10 nodes, vs size of the graph (in terms of number of nodes N , plotted on logarithmic scale) for generalized PBOs (Kuramoto model) (green plot), spintronic oscillators (Slavin's model) (red plot), and GW algorithm (blue plot) for three kinds of graphs: (a) Mobius Ladder, (b) random cubic, (c) and Erdős Rényi. No thermal noise has been incorporated in either Kuramoto model or Slavin's model to obtain this plot.

Figure 9 shows that mean AR for the case with noise almost matches with that for the case without noise for almost all graph instances. Figure 10 shows that best case AR numbers are also comparable for most graph instances, with best case AR for the noisy case sometimes exceeding that for the case without noise. Similar observation has also been made

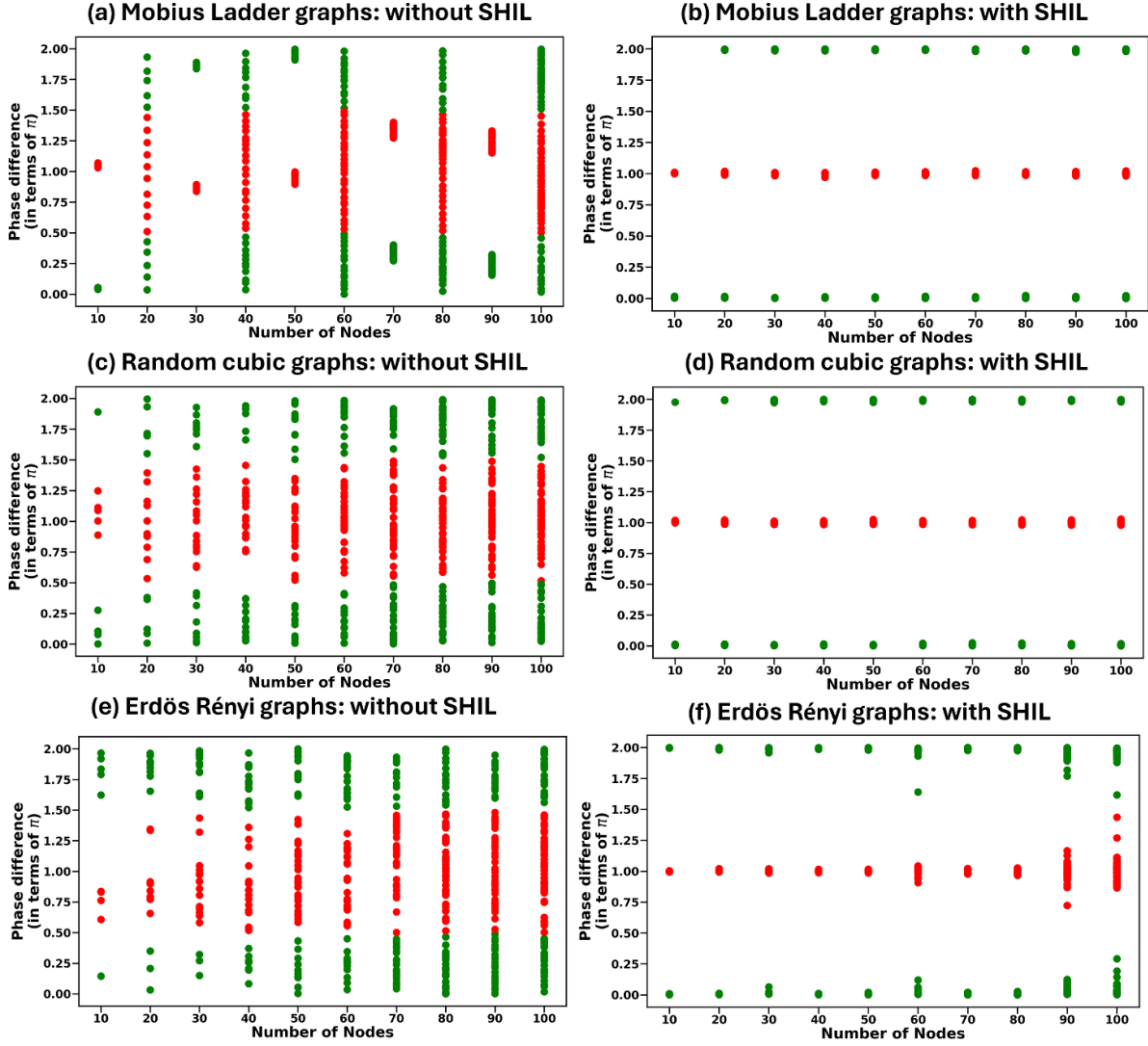


Figure 8. Final steady state phase difference of any oscillator with respect to the reference oscillator is plotted as a circle/dot for every graph instance, as obtained from the dynamics predicted by the Kuramoto model of oscillators for that graph instance. As graph size increases, the number of dots increases since the number of oscillators increases. For (a), (c), and (e), the SHIL term is absent in the Kuramoto model (equation (2): $K_s = 0$), but for (b), (d), and (f), the SHIL term is present with $K_s = 0.5 \text{ radian s}^{-1}$.

in the reports by Wang *et al* [13, 14], with the explanation provided there being that noise can sometimes help the system to get out of local minima in the higher dimensional space across which the system is optimizing. Figures S1 and S2 of section S1 of supplementary material show that the same logarithmic growth of TTS with graph size for oscillators, which we observed for noiseless Kuramoto model earlier, is maintained even in the presence of noise. Thus as per Kuramoto model, thermal noise does not adversely affect the accuracy and speed of OIMs, which is a major advantage for OIMs.

Next, similar to the method we followed above for adding noise to Kuramoto model, we have included noise in the Slavin's model of spin oscillators: we add an extra term $\mathcal{N}(0, 1)$ (with a different pre-factor here though (100 MHz)

from Kuramoto model since the operating frequency is different) to the right hand sides of equations (4) and (5). The same method to incorporate noise in Slavin's model of spin oscillators has been followed in a report by Grimaldi *et al* [24]. For the graph instances listed in table 2, we have carried out one trial for Slavin's model without noise and one trial for Slavin's model with noise incorporated. The corresponding phase evolution plots are shown in section S2 of supplementary information accompanying the paper (figures S3–S12). We observe that while noise indeed changes the phase evolution trajectory (as expected), the final cut scores match between noiseless and noisy case for 7 out of the 10 graph instances we have explored (ML graphs with 10, 20, 30, 40, and 50 nodes, and RC graphs with 10 and 20 nodes). For 2 graph instances (RC graphs

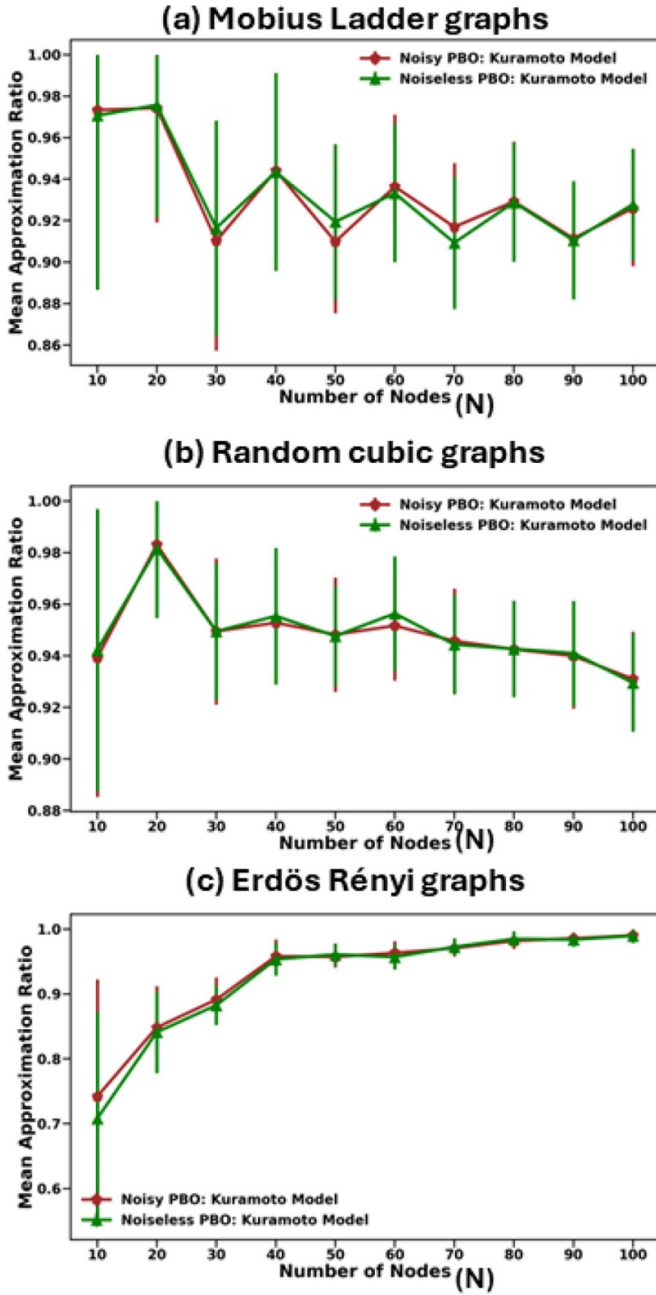


Figure 9. Mean approximation ratio vs size of the graph (in terms of number of nodes N) for generalized PBOs without incorporating noise in Kuramoto model (green plot) and after including noise in Kuramoto model (red plot) and three kinds of graphs: (a) Mobius Ladder, (b) random cubic, (c) and Erdős Rényi. Standard deviation of cut values (divided by best known Max-Cut value) is shown as an error bar around the mean value.

with 30 and 50 nodes), noiseless simulation yields higher cut value than noise-included one. But for 1 graph instance (40-node RC graph), inclusion of noise yields a higher cut value than the noiseless case. This finding is consistent with that obtained from Kuramoto model as described above: noise can sometimes help in escaping local minima in the optimization landscape [13, 14]. Figures S3–S12 of supplementary material also shows that noise in Slavín's model does not affect the TTS

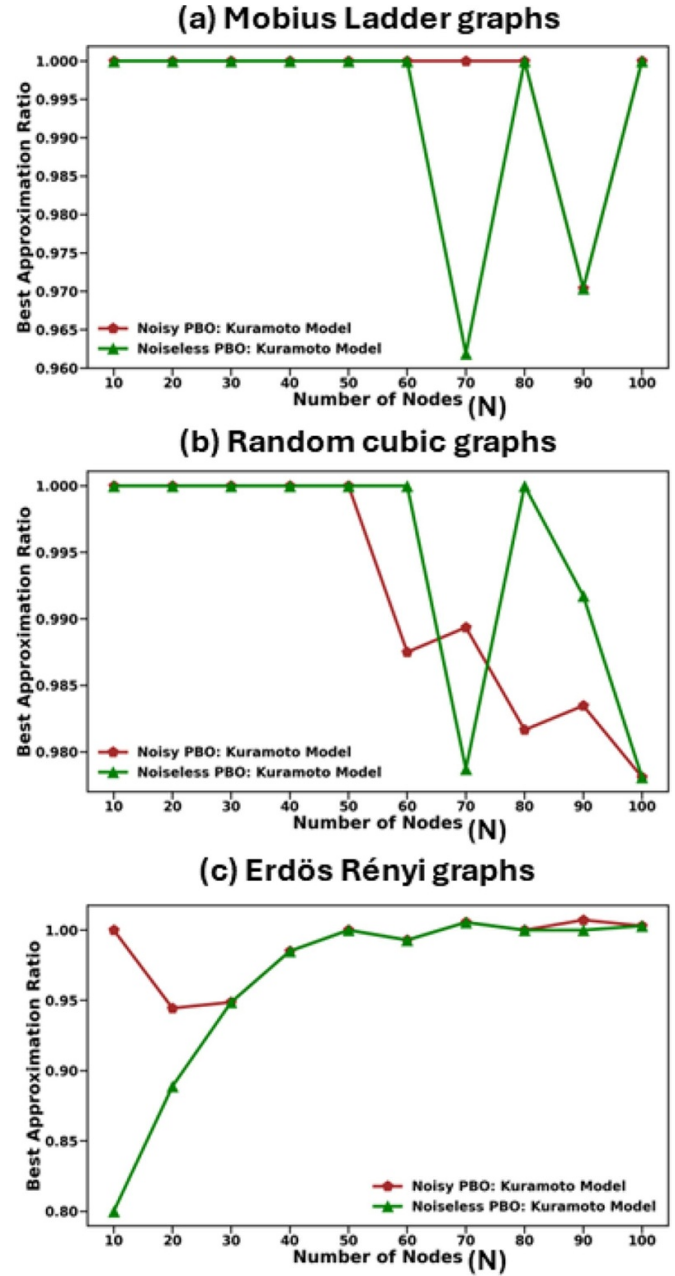


Figure 10. Best case approximation ratio vs size of the graph (in terms of number of nodes N) for generalized PBOs without incorporating noise in Kuramoto model (green plot) and after including noise in Kuramoto model (red plot) and three kinds of graphs: (a) Mobius Ladder, (b) random cubic, (c) and Erdős Rényi.

much either since the phases of the oscillators reach steady state pretty much around the same time, with or without noise.

5.3. Dependence on graph sparsity

It is to be noted that both ML and RC graphs have an edge connectivity of 3, i.e. each graph node is connected to only three other nodes. Even the ER graphs we have explored here belong to $G(N, 0.1)$ category (as mentioned before), i.e. only 10% of all possible edges exists in any graph instance. Thus, the graphs we have explored are sparse in nature. Other recent

Table 2. Comparison of cut values obtained from Slavin's model of spin oscillators (without and with noise incorporated in them) for different cut values. The corresponding phase evolution plots are shown in section S2 of supplementary material (figures S3–S12).

Graph Instance	Cut value: Slavin's model noiseless	Cut value: Slavin's model with noise
10-node ML	15	15
20-node ML	28	28
30-node ML	45	45
40-node ML	58	58
50-node ML	75	75
10-node RC	11	11
20-node RC	25	25
30-node RC	35	34
40-node RC	48	49
50-node RC	62	59

reports on spintronic OIMs also explore such sparse graphs primarily [35]. However, there are some recent reports on electronic-ring-oscillator-based Ising machines which solve Max-Cut on graphs of fairly high edge density [18]. Exploring Max-Cut on such dense graphs with spintronic OIMs will be our future work.

5.4. Nature of coupling between spintronic oscillators

In this paper, we have used the Slavin's model of spin oscillators to model the spintronic OIM. In the Slavin's model (given by equations (4) and (5)), coupling between two spintronic oscillators j and j' is given by the coupling coefficient $\Omega_{jj'}$, which is agnostic of the nature of the coupling (dipole, spin wave, or electrical). Thus, our results in the paper are applicable to any form of coupling that is ultimately used for experimental implementation of the spintronic OIM.

However, in this context, it is to be noted that implementing dipole coupling and spin-wave coupling has associated challenges in the context of OIMs because these are global coupling mechanisms and the strength of the coupling depends on the physical distance between the spin oscillators. So, it is hard to modify the coupling strengths as one switches from solving one graph instance to another on the spintronic OIM.

Though spin-wave coupling has this challenge, currently research is being pursued actively to experimentally demonstrate Ising computing in spin-wave-coupled spin Hall nano-oscillator (SHNO) systems [9]. It has also been recently shown experimentally that in the same spin-wave-coupled nano-constriction SHNO system, synchronization between oscillators can be controlled by memristive gating [44], which can enable tuning of coupling coefficients needed for OIMs.

Electrical coupling is another possible way of implementing coupling in spintronic OIMs. Unlike global coupling mechanisms like dipole coupling and spin-wave coupling, electrical coupling is a local coupling mechanism, and coupling between any two oscillators can be individually controlled without affecting the coupling between other oscillators. Electrical coupling between spin torque oscillators, which

operate in the uniform precession mode and are not vortex-based, has been demonstrated experimentally in the 2.4 GHz range in a recent report by Sharma *et al* [29].

6. Conclusion

Thus, in this paper, we show that for the Max-Cut problem solved on three kinds of graphs (ML, RC, ER) up to 100 nodes, TTS for PBOs, based on Kuramoto model, grows at a much lower rate (logarithmically: $\mathcal{O}(\log(N))$) with respect to graph size N , compared to conventional state-of-the-art classical approximation approximations like GW algorithm ($\mathcal{O}(N^2)$), meant to be implemented on conventional classical digital computers (figure 6), while maintaining the same level of accuracy (figures 4 and 5). This, however, is a general trend and is device-physics agnostic. If different oscillator technologies have phase dynamics that follow Kuramoto model, then their TTS will grow logarithmically with graph size (figure 6).

Compared to other oscillator technologies, spintronic oscillators operate at a higher frequency (in GHz). So, TTS for spintronic oscillators on graphs of 10 nodes (listed in table 1) is lower compared to that for other oscillator technologies, and hence, TTS for spintronic OIMs will be proportionately lower than that of other OIMs [3, 15, 16] for large graphs. Our modeling of the physics of coupled magnetization dynamics in spintronic oscillators through Slavin's model further shows that TTS of spintronic oscillators also grows logarithmically ($\mathcal{O}(\log(N))$) with graph size N , while their accuracy is comparable to that of GW. So spintronic oscillators have improved time complexity over GW algorithm, and for large graphs, they are expected to compute Max-Cut values much faster than GW algorithm, as well as other oscillators that operate at lower frequencies, while maintaining the same level of accuracy. We have also incorporated noise in our oscillator models and found this result to be true even in the presence of noise.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

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References

- [1] Chvatal V 2011 *Combinatorial Optimization: Methods and Applications* (IOS Press)
- [2] Unger R and Moulton J 1993 *Bull. Math. Biol.* **55** 1183–95
- [3] Ahmed I, Chiu P, Moy W and Kim C 2021 *IEEE J. Solid-State Circuits* **56** 2870–80
- [4] Sipser M 2013 *Introduction to the Theory of Computing* (Cengage Press)
- [5] Goemans M X and Williamson D P 1995 *J. Assoc. Comput. Mach.* **42** 1115–45
- [6] Gabow H N, Goemans M X and Williamson D P 1998 *Math. Program.* **82** 13–40
- [7] Lucas A 2014 *Front. Phys.* **2** 5
- [8] Mohseni N, McMahon P L and Byrnes T 2022 *Nat. Rev. Phys.* **4** 363–79
- [9] Houshang A, Zahedinejad M, Muralidhar S, Chechinski J, Khymyn R, Rajabali M, Fulara H, Awad A A, Dvornik M and Åkerman J 2022 *Phys. Rev. Appl.* **17** 014003
- [10] Haribara Y S U and Yamamoto Y 2016 *Principles and Methods of Quantum Information Technologies, Lecture Notes in Physics*, ed Y Yamamoto and K Semba (Springer) p 911
- [11] Hamerly R et al 2019 *Sci. Adv.* **5** eaau0823
- [12] Zhang Y et al 2022 *Micromachines* **13** 1016
- [13] Wang T and Roychowdhury J 2019 Oim: oscillator-based ising machines for solving combinatorial optimisation problems *Proc. Int. Conf. on Unconventional Computation and Natural Computation*
- [14] Wang T 2020 Novel computing paradigms using oscillators *PhD dissertation* University of California Berkeley Department of Electrical Engineering and Computer Sciences
- [15] Delacour C, Carapezzi S, Boschetto G, Abernot M, Gil T, Azemard N and Todri-Sanial A 2023 *Neuromorph. Comput. Eng.* **3** 034004
- [16] Wang T, Wu L and Roychowdhury J 2019 New computational results and hardware prototypes for oscillator-based ising machines *Proc. 56th Annual Design Automation Conf. (DAC) 2019 (Las Vegas, NV, USA)* p 239
- [17] Moy W, Ahmed I, Chiu P W, Moy J, Sapatnekar S S and Kim C H 2022 *Nat. Electron.* **5** 310–7
- [18] Lo H, Moy W, Yu H, Sapatnekar S and Kim C H 2023 *Nat. Electron.* **6** 771–8
- [19] Mallick A, Bashar M K, Truesdell D S, Calhoun B H, Joshi S and Shukla N 2020 *Nat. Commun.* **11** 4689
- [20] Shukla N et al 2014 *Sci. Rep.* **4** 4964
- [21] Dutta S, Khanna A, Assoa A S, Paik H, Schlom D, Toroczka Z, Raychowdhury A and Datta S 2023 *Nat. Electron.* **4** 502–12
- [22] Zahedinejad M, Awad A A, Muralidhar S, Khymyn R, Fulara H, Mazraati H, Dvornik M and Åkerman J 2020 *Nat. Nanotechnol.* **15** 47–52
- [23] Finocchio G, Ventra M D, Camsari K Y, Everschor-Sitte K, Amiri P K and Zeng Z 2021 *J. Magn. Magn. Mater.* **521** 167506
- [24] Grimaldi A, Mazza L, Raimondo E, Tullo P, Rodrigues D, Camsari K Y, Crupi V, Carpentieri M, Puliafito V and Finocchio G 2023 *Phys. Rev. Appl.* **20** 024005
- [25] Garg N, Bhotla S V H, Muduli P K and Bhowmik D 2021 *Neuromorph. Comput. Eng.* **1** 024005
- [26] Bhotla S V H, Garg N, Aggarwal T, Muduli P K and Bhowmik D 2023 *IEEE Trans. Nano* **22** 136–48
- [27] Singh H, Bhuktare S, Bose A, Fukushima A, Yakushiji K, Yuasa S, Kubota H and Tulapurkar A A 2019 *Phys. Rev. Appl.* **5** 054028
- [28] McGoldrick B C, Sun J Z and Liu L 2022 *Phys. Rev. Appl.* **17** 014006
- [29] Sharma R, Mishra R, Ngo T, Guo Y X, Fukami S, Sato H, Ohno H and Yang H 2021 *Nat. Commun.* **12** 2924
- [30] McSorley J P 1998 *Discrete Math.* **184** 137–64
- [31] Noy M, Requilé C and Rué J 2019 *Random Struct. Algorithms* **56** 892–924
- [32] Erdős P and Rényi A 1959 *Publ. Math. Debrecen*
- [33] Kuramoto Y 2003 *Chemical Oscillations, Waves and Turbulence* vol 19 (Springer)
- [34] Slavov A and Tiberkevich V 2009 *IEEE Trans. Magn.* **45** 1875–918
- [35] Albertsson D I, Zahedinejad M, Houshang A, Khymyn R, Åkerman J and Rusu A 2021 *Appl. Phys. Lett.* **118** 112404
- [36] Chen T, Eklund A, Iacocca E, Rodriguez S, Malm B G, Åkerman J and Rusu A 2015 *IEEE Trans. Electron Devices* **62** 1037–44
- [37] Hagberg A A, Schult D A and Swart P J 2008 Exploring network structure, dynamics and function using networkx *Proc. 7th Python in Science Conf. (Pasadena, CA USA)* pp 11–15
- [38] Kim J H and Vu V H 2003 Generating random regular graphs *Proc. Association for Computing Machinery (ACM) (New York, NY, USA)* pp 213–22
- [39] Andersen M S et al 2013 54 (available at: cvxopt.org)
- [40] Virtanen P et al 2020 *Nat. Methods* **17** 261–72
- [41] Kalinin K P and Berloff N G 2022 *Commun. Phys.* **5** 20
- [42] Georges B, Grollier J, Cros V and Fert A 2008 *Appl. Phys. Lett.* **92** 232504
- [43] Flovik V F M and Wahlström E 2016 *Sci. Rep.* **6** 32528
- [44] Zahedinejad M, Fulara H, Khymyn R, Houshang A, Dvornik M, Fukami S, Kanai S, Ohno H and Åkerman J 2022 *Nat. Mater.* **21** 81–87